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BIOMECHANICS OF PRONATION AND SUPINATION OF THE FOREARM

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Pronation–supination is the motion of rotation of the forearm around its longitudinal axis. It is an important motion because it allows the hand to be oriented upwards—supination—and downwards—pronation.[1] This motion is added to those of the wrist, flexion–extension, and radial and ulnar inclinations, and makes it a three-degrees-of-freedom joint. With this possibility of orientation (Fig. 1), the hand can take food from a table and carry it to the mouth—the vital function of feeding—and, thanks to pronation, it is possible to lean against an obstacle and overcome it. Supination is necessary to support heavy loads or pick up change at the supermarket. Supination is also indispensable to perform personal hygiene of the posterior perineum, a necessary condition for individual autonomy.

THE TWO BONES OF THE FOREARM

A question arises concerning the function of pronation–supination: Why are there two bones in the forearm?[2] In other words, why isn't there one bone alone in the forearm and a ball-and-socket joint at the wrist? If that were the case, the tendons (Fig. 2), nerves and arteries (Fig. 3) would be twisted at the level of the wrist, causing serious difficulties of functioning. To avoid muscular problems, one can imagine them installed in the hand, but then, the hand would be monstrous, convex and heavy instead of concave, slim and light. On the contrary, with two bones turning around themselves, it is possible to set muscles, nerves and vessels on the turning bone—the radius—without any risk of twisting. So the ulna—the steady bone—is articulated with the humerus, through the elbow, and the rotating radius supports the hand through the wrist.

We inherited this disposition with two bones at the forearm and the leg from a remote ancestor, the ichtyostega (Fig. 4), a crossopterygian tetrapode, who sprang up from the sea 300 million years ago, transforming its fins into members.

THE ROTATION OF THE RADIUS AROUND THE ULNA

Thanks to two joints uniting the radius and the ulna, the proximal radioulnar joint (PRUJ) and the distal radioulnar joint (DRUJ), the one can rotate around the other. These two bones are like a rectangular frame[3] with an oblique hinge on the diagonal (Fig. 5). When this frame is flat, the two bones are parallel and aside, with the radius lateral to the ulna. When the radius rotates around the hinge, with a 180° amplitude, it crosses over the

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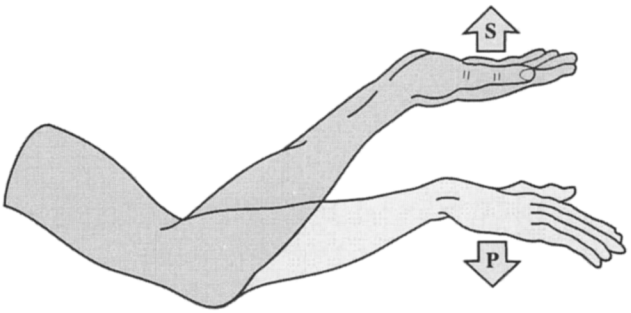


Figure 1. The usefulness of the pronation–supination motion. Supination (S) is necessary to support heavy loads or pick up change at the supermarket. Pronation (P) allows a person to lean against an obstacle or take food on a table and carry it to the mouth.

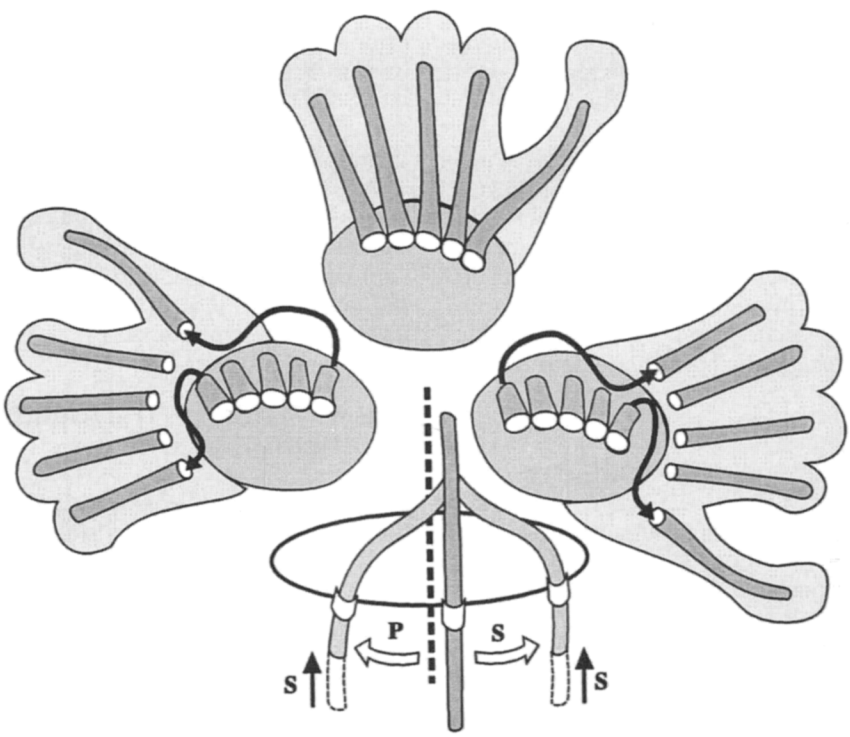


Figure 2. The tendinous problem with a hypothetic ball and socket joint at the wrist. The tendons are twisted at the level of the wrist and their length would be changed during the pronation–supination motion. S = supination; P = pronation.

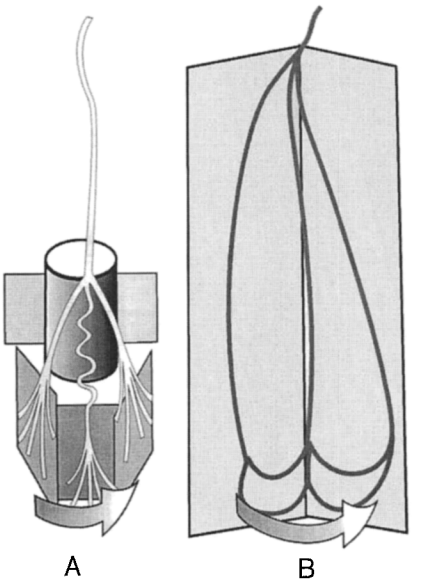


Figure 3. The vascular problem with a hypothetic ball and socket joint at the wrist. *A*, The arteries are twisted with a change of length. *B*, With a longitudinal hinge, the twisting is distributed on the whole length of the forearm.

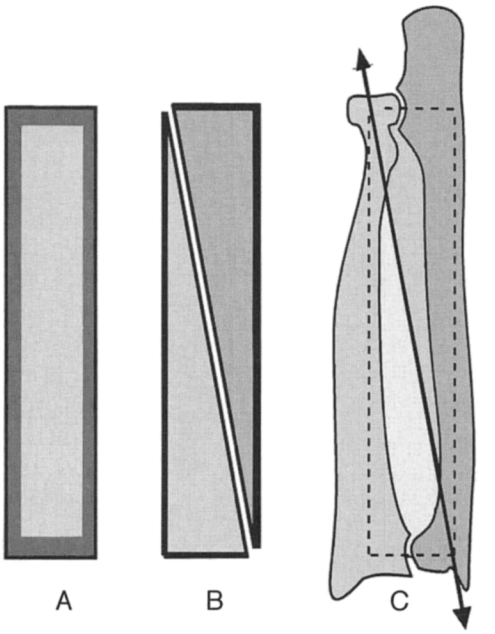


Figure 5. The skeleton of the forearm. It is considered a rectangular frame *(A)* with an oblique hinge on the diagonal *(B)*. When this frame is flat *(C)*, the two bones are parallel and aside the radius lateral regarding to the ulna.

ulna, at the front of it. The result would be a changing of direction of the lower joint—the wrist. If this obliquity were not compensated by the inverse obliquity of the upper joint—the elbow, that has a built in disposition, the physiologic cubitus valgus (Fig. 6). So, the axis of the hand remains in the continuation of the longitudinal axis of the forearm.

THE CONDITIONS OF A FREE MOTION

The axes of the two radioulnar joints must be aligned coaxially; if not, there is a blocking: When the two hinges of a door are not aligned, the door cannot be opened (Fig. 7). This ideal line joining the center of the radioulnar joints can be considered as the mechanical axis of prono-supination. These joints are of trochoid type, with roughly cylindrical surfaces. The PRUJ is integrated into the elbow joint, whereas the DRUJ belongs to the wrist.

THE STABILITY OF THE PROXIMAL RADIOULNAR JOINT

The stability of the PRUJ is excellent (Fig. 8) because the radial head is enclosed in a very strong osteoligamentous cavity, made of the radial notch of the ulna and the annular ligament, which is a very resistant ligament; it is also deformable so as to conform itself to the slight ovality of the radial head. There are rarely problems with that joint, except in fracture of the radial head.



Figure 4. Our remote tetrapode ancestor. The ichtyostega, a crossopterygian tetrapode, who sprang up from the sea 300 million years ago, transforming its fins into members, had two bones at the forearm and the leg.

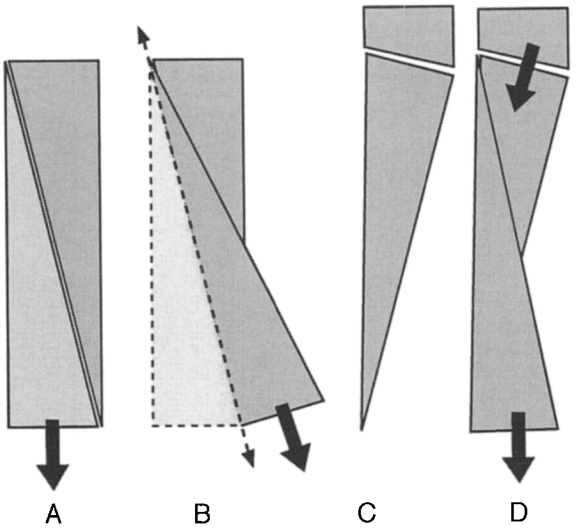


Figure 6. The usefulness of the cubitus valgus. With an oblique hinge *(A)*, the rotation of the radius *(B)* changes the orientation of the wrist. With the cubitus valgus *(C)* caused by the obliquity of the elbow axis, the direction of the wrist and the hand *(D)* remains longitudinal.

THE STABILITY OF THE DISTAL RADIOULNAR JOINT

The DRUJ is much more fragile because its surfaces are not congruent (Fig. 9), the radial sigmoid notch being less wide than the development of the head surface, and because the curvature of radius is not equal to that of the ulnar head. But the main reason for its potential instability is that its linkage system is vulnerable. Actually, the main ligament is the triangular ligament, which also makes the separation between the radiocarpal joint and the radioulnar joint like a meniscus. It is biconcave, with two articular aspects and a thin central part, sometimes perforated, and two reinforcements along its anterior and its posterior borders. The base of this triangle is inserted on the inferior ridge of the radial sigmoid notch, and its vertex at a point located between the base of the ulnar styloid process and the lower surface of the ulnar head. Under pressing and splitting strains,

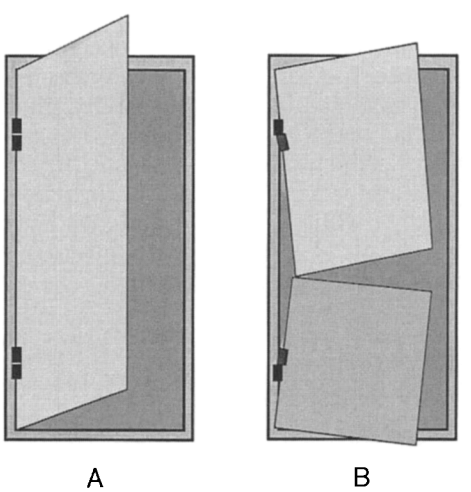


Figure 7. Coincidence of the axes of the radioulnar joints. *A*, The axes of the two radioulnar joints must be aligned, coaxial, like the hinges of a door. *B*, If not, there is a blocking and the door cannot be opened.

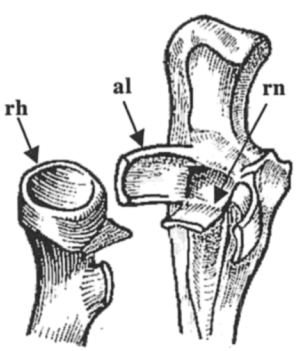


Figure 8. The proximal radioulnar joint. Excellent stability: the radial head (rh) is enclosed in a strong osteoligamentous cavity made of the radial notch (rn) of the ulna and of the annular ligament (al).

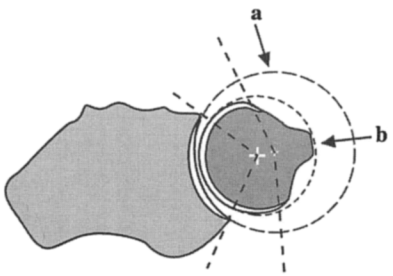


Figure 9. The incongruency of the joint surfaces of the distal radioulnar joint (DRUJ). The curvature of radial notch *(a)* is not equal to that of the ulnar head *(b)*. Therefore, the joint surfaces are incongruent and the centers of rotation are not coincidental.

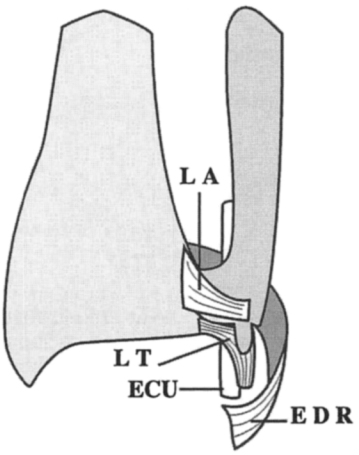


Figure 10. Reinforcement of the triangular ligament. The triangular ligament (LT) is reinforced by the anterior ligament (L.A.) of the DRUJ, the ECU tendon comprised in its sheath, and the medial expansion of the dorsal retinaculum (EDR).

this ligament is often torn or de-inserted. However, it is reinforced with other ligamentous structures (Fig. 10)—the anterior and posterior ligaments of the DRUJ, a medial expansion of the dorsal retinaculum of the wrist, and particularly the tendon of the extensor carpi ulnaris (ECU) in its sheath, which is a mighty reinforcement. Because of the structure of the triangular ligament (Fig. 11), the tension of each band is equal in intermediate position, but with the insertion of the vertex being medial with regard to the center of rotation of the ulnar head, it is obvious that the posterior band is tightened in pronation and the anterior one in supination. The action of these bands is to center the ulnar head, which is not quite steady in the radial notch during prono-supination.

The motion of the DRUJ is a complex one (Fig. 12): The radial epiphysis turns around the ulnar head, whereas this one is doing a translation on a circular path, without any rotation on itself. This motion combines extension and lateral translation. The result is a ballet around an instantaneous mobile axis located somewhere in the radial epiphysis, not in the ulnar head. Rotation is not the only movement of the ulnar head in the radial notch. In fact, the progressive obliquity of the radius, crossing over the ulna during pronation, causes an inclination with regard to the radial notch; the reason why the joint surface of the head is slightly convex vertically, but the obliquity of this vertical line is variable among individuals. Moreover, in the pronation position (Fig. 13), the radius becomes

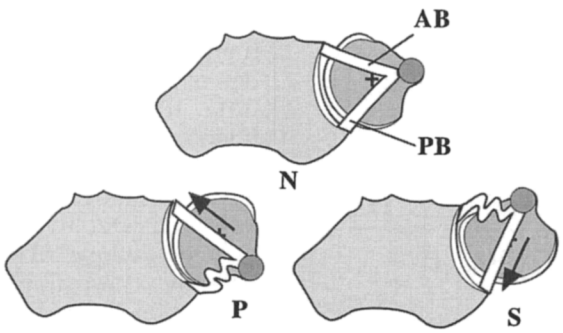


Figure 11. Structure and functioning of the triangular ligament *(inferior view)*. In neutral position (N), the reinforcing bands of the triangular ligament, the anterior (AB) and the posterior (PB), are equally stretched. In pronation (P) the anterior band is stretched, whereas in supination (S) the posterior is tightened.

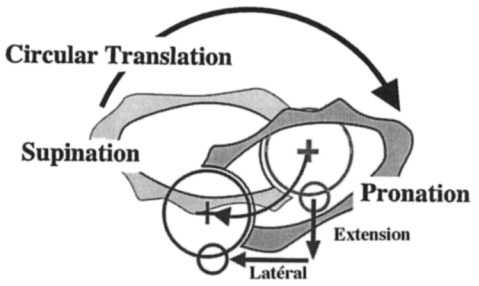


Figure 12. The ballet of the radius around the ulna. From the supination to the pronation, the radius is rotating around the ulnar head. The ulnar head is moving along a circular path without any translation, combining extension and lateral translation.

shorter than the ulna and the inferior aspect of the ulnar head moves downward with regard to the lower rim of the radial notch. When, after a distal radius fracture, the ulnar variance (UV) is already positive, pronation increases the impingement of the ulnar head on the carpus bones—lunate or triquetrum.

THE RANGE OF THE PRONATION–SUPINATION MOTION

The total range of pronation–supination (Fig. 14) is roughly 180°, but the motion must be measured from a reference position (*O*)—elbow flexed at right angle and hand included in a vertical plane, the thumb being directed upward. From this position, the pronation (*P*) is 90°, rotating the palm downward and the thumb medially. In the inverse motion–supination–the palm is directed upward and the thumb is directed laterally. With the

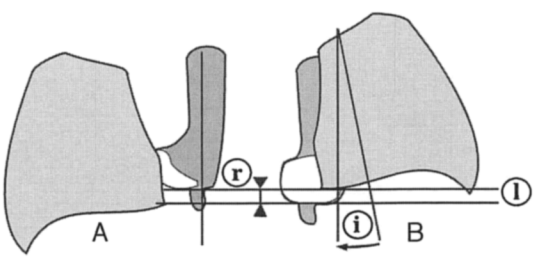


Figure 13. The variations of the ulnar variance (UV). *A*, In supination the two bones are parallel and the UV is negative (r). *B*, In pronation, because of the fact that it becomes oblique (i), the radius is relatively shorter than the ulna, and the UV becomes positive. The pronation increases the impingement of the ulnar head on the carpus bones: lunate or triquetrum, when there is a preexisting shortening of the radius.

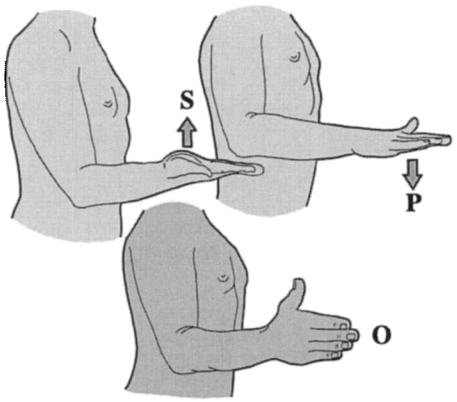


Figure 14. The amplitude of the pronation–supination motion. This motion is measured from the zero position (O), elbow 90° flexed and the thumb directed upwards. The pronation (P) is 85° to 90° when the thumb is directed inwards and the palm faces downwards. The supination is 90° when the thumb is directed laterally and the palm faces upwards (S).

"waiter's test" (Fig. 15), it is possible to know whether supination is complete: A tray carrying a glass of water can be held upon the palm without pouring the liquid. Supination, normally 90°, is the most important movement because it cannot be compensated by the shoulder, whereas a lack of pronation can be replaced by its abduction.

THE COAPTATION BETWEEN RADIUS AND ULNA

The solidarity between the two bones of the forearm is assumed not only by the ligaments of the two radioulnar joints, but also by the interosseous membrane and the muscles.

The interosseous membrane (Fig. 16) is actually a flexible hinge joining the medial border of the radius to the lateral border of the ulna. It is made of two layers of oblique fibers—one anterior, mainly represented by fibers running obliquely distally and medially, and some fibers obliquely directed distally and laterally; one posterior layer made only of oblique fibers running distally and laterally. These fibers play an essential role in binding the radius and ulna together and stopping the slippage of the radius proximally for the anterior layer and distally for the posterior one. Upward, the radius is also blocked by the humeral condyle, but downward, the posterior layer plays an important role, helped by the oblique cord of Weit-

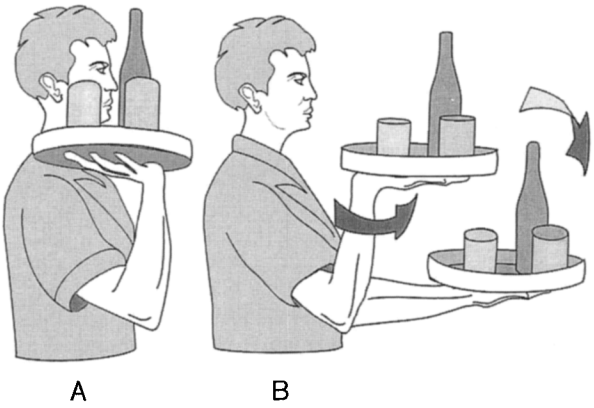


Figure 15. The waiter's test. From the walking position *(A)*, in which the tray is held at the shoulder, to the serving position *(B)*, in which the waiter's forearm executes a supination motion. When the supination is complete, the tray is horizontal and the liquid in the glasses is not poured out.

brecht, joining the extremities of the two bones. However, the main factor that prevents dislocation of the elbow and the wrist, and slippage of the radius from the ulna is the fact that the flexor muscles of the fingers (Fig. 17) are bi-articular; they are partly attached to the humeral lower extremity so that when they are in contraction to support a load in the hand, they drive the radius upward beside the ulna.

Because of the insertion of the flexor muscles on the anterior aspect of the membrane, they are compressed between the two bones during pronation (Fig. 18). That makes a soft abutment of the motion and tightens the membrane. An important and active link between the extremities of the two bones is the pronator quadratus, which acts as an active ligament during the pronation.

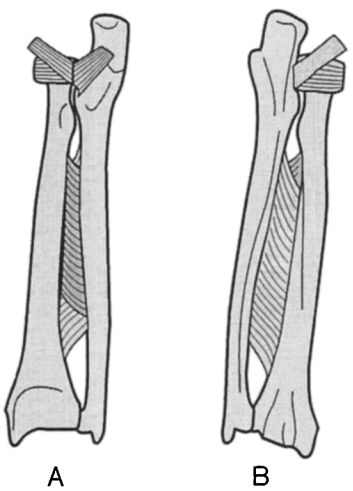


Figure 16. The interosseous membrane. This flexible hinge is made of two layers of oblique fibers: one anterior *(A)*, mainly represented by fibers running obliquely, distally, and medially, and one posterior *(B)* made only of oblique fibers running distally and laterally.

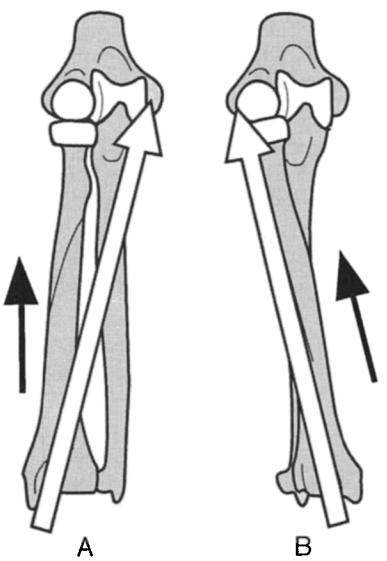


Figure 17. The longitudinal coaptation. The flexor muscles of the fingers are biarticular; they are attached to the humeral lower extremity so when they are in contraction to support a load with the hand, they drive up the radius aside the ulna, either in supination *(A)* or in pronation *(B)*.

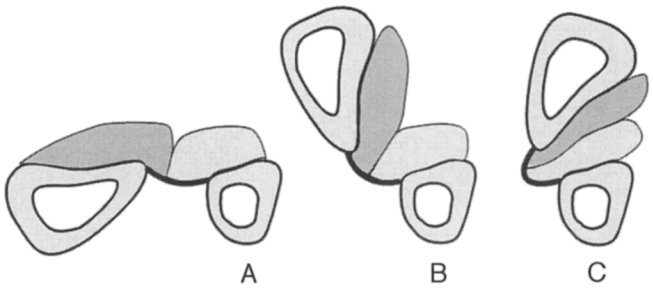


Figure 18. The soft abutment of the two bones in pronation. Because of the insertion of the flexor muscles on the anterior aspect of the membrane *(A)*, they are compressed between the two bones during pronation *(B)*, making a soft abutment of the motion and tightening the membrane *(C)*.

THE MOTOR MUSCLES OF PRONATION–SUPINATION

There are two pairs of motor muscles for pronation–supination. They are fixed on precise points on the radius, the shape of which is roughly crank-like, with two extremities and two angles. The supinator muscle is fixed on the upper end of the radius—the supinator brevis. It acts by a mechanism of unrolling. The second muscle fastens into the radial tuberosity at upper angle of the "crank"—the biceps—a very mighty muscle that is also a flexor of the elbow. The brachioradialis is not usually a supinator, it only is one when the forearm is in complete pronation. Actually, it is a flexor of the elbow. The pronator muscles have a symmetric disposition: One, the pronator teres, inserts into the apex of the pronator bend of the radius; its action is weak especially when the elbow is extended. The second one, the pronator quadratus, wraps around the distal end of the ulna, acting by "unwinding." On the whole, the pronator muscles are less powerful than the supinators, and the efficiency of these muscles depends on the intact shape of this "crank." A malunion after a radius fracture therefore may compromise this mechanism.

The question regarding the pronation–supination motion is: How is the torque of this movement transmitted to the hand? In fact, the condyle of the carpus is not firmly embedded in the radial glena, and when the hand is fixed on an object, the radiocarpal joint can be dislocated under the strain of pronation–supination. With CT scan[5] with and without strain of pronation–supination (Fig. 19), it has been possible to establish that the rotational shift of the carpus below the radius is 45° when the hand muscles are not contracted and only 15° when they are. The "caging" of the muscles around the wrist is a perfect protection against dislocation, and when hand muscles are contracted, they act as a "clutch mechanism."

TROUBLES IN PRONATION–SUPINATION AND HOW TO CURE THEM

To summarize this discussion, pronation–supination is a sophisticated and precise mechanism, and there are many factors that may lead to problems. The author shall consider only the traumatic causes, which are seen most frequently.

The isolated fractures of one of the two bones, or the fracture of both bones, by changing the precise geometry of the frame, may compromise definitively the range of prono-supination, because the curvature of one of them is inversed and causes an abutment. Treatment consists of the restoration of the normal shape of the bones with appropriate osteotomy. Sometimes, the fracture of one bone is associated with a luxation of one of the two joints:

- Fracture of the ulna combined with an anterior luxation of the PRUJ (Monteggia's fracture; Fig. 20*A*). The diagnosis requires a radiograph of the whole forearm to avoid missing the dislocation of the upper joint.
- Fracture of the radius associated with an anterior luxation of the DRUJ (Galeazzi's fracture; Fig. 20*B*). The diagnosis problem is the same: Do not miss the joint dislocation.

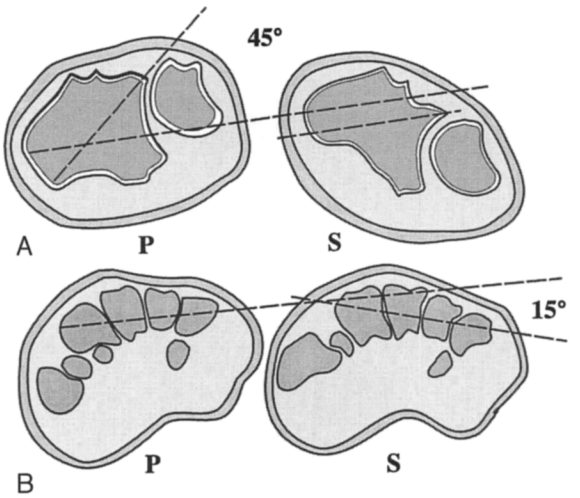


Figure 19. The transmission of the torque of pronation–supination to the hand. With CT scan under and without costrain of pronation (P) and supination (S), the rotatory shifting of the carpus at the DRUJ level *(A)* is 45° and only 15° at the metacarpal level *(B)* when the hand muscles are contracted.

These two types of fractures can cause severe problems in pronation–supination if the fracture and the luxation are not correctly fixed.

Occurring on a congenital malformation—the Madelung illness—a simple wrist sprain can provide definitive problems in pronation–supination and may require a difficult operation.

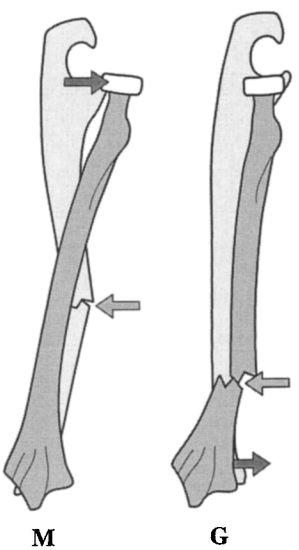


Figure 20. Fractures of one bone combined with a luxation of the other. The Monteggia's (M) fracture combines a fracture of the ulna with a luxation of the proximal radioulnar joint (PRVJ). Galeazzi (G) fracture combines a fracture of the radius with a dislocation of the DRUJ.

A fracture of the radial head sometimes may cause, by way of a radial shortening, a dislocation of the DRUJ. It is named the *Essex-Lopresti syndrome* (Fig. 21), and is difficult to fix. It can also occur after removing the radial head following a fracture.

Isolated dislocations of the DRUJ are rare, caused by an extreme motion of pronation for the posterior dislocation, or a forced movement of supination for the anterior luxation. The treatment is difficult and requires a type of operation discussed subsequently.

The most frequent dislocation of the DRUJ is caused by the fracture of the distal radius. The tilt of the radius's lower epiphysis produces four types of disorders (Fig. 22):

- a widening of the joint, by its lateral shift
- a change of the axis of the sigmoid notch because of posterior tilt
- a lifting up of the lower border of the sigmoid notch
- an impingement of the ulnar head on the lunate

These criteria can be noticed on plain radiograph using anteroposterior and lateral views. Consequently, pain appears on the medial side of the wrist and pronation–supination amplitudes are limited. There are

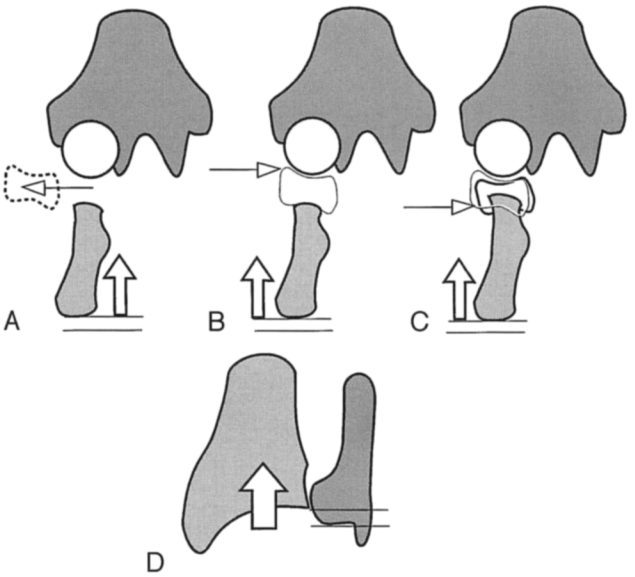


Figure 21. The Essex–Lopresti syndrome. Three causes of shortening of the radius: avulsion of the ulnar head *(A)*, wearing of the upper aspect of the ulnar head *(B)*, and impaction of the ulnar head *(C)* generate a positive UV and an impingement ulno carpal *(D)*.

many ways to treat these problems, according to the importance of the impingement and the integrity of the joint surfaces and ligaments of the DRUJ.

For the treatment of a painful impingement of the ulnar head on the lunate, several procedures are available, according to the integrity of the triangular ligament: If this ligament is preserved, it is possible to make an ulnar shortening (Milch procedure) using either a

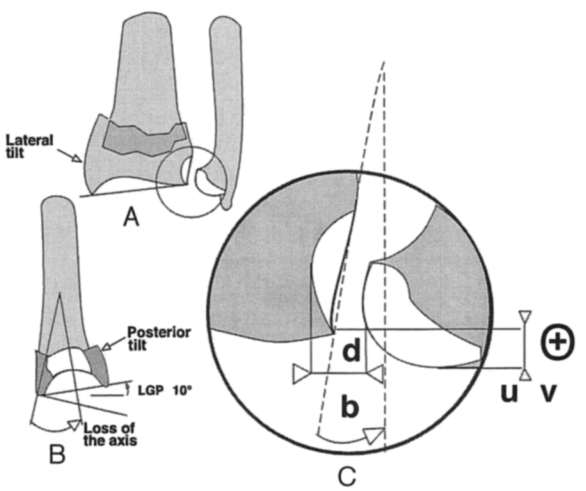


Figure 22. The dislocation of the DRUJ caused by the fracture of the distal radius. The lateral tilt *(A)* and the posterior tilt *(B)* cause the inversion of the glenoid line (LGP) and dislocate the DRUJ *(C)*; d = a widening of the joint; b = a change of the axis of the sigmoid notch; uv+ = a lifting up of the lower border of the sigmoid notch, causing an impingement of the ulnar head on the lunate.

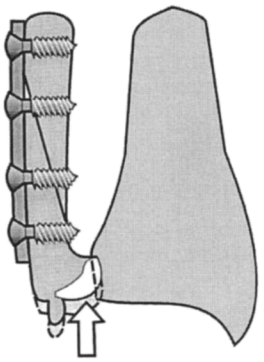


Figure 23. The Milch procedure. The painful impingement of the ulnar head on the lunate may be treated with a shortening oblique osteotomy of the ulna.

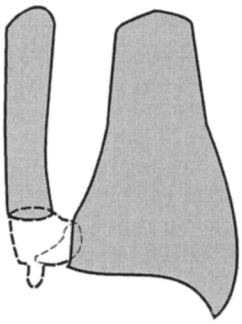


Figure 25. The Moore–Darrach procedure. The avulsion of the ulnar head must be the most economic as possible and save the maximum of ligament connections.

short oblique cut with compression or a long oblique resection with a 4 or 6 hole plate (Fig. 23). It is a risky operation with a high rate of complications because the ulnar consolidation is long and hazardous.

There are also several procedures of shortening and remodeling the ulnar head (Fig. 24), either with an open technique—Bowers or K. Watson—or with an arthroscopic Waffer procedure.

When the triangular ligament is destroyed, a procedure often used is the Moore-Darrach procedure. This involves resection of the ulnar head (Fig. 25) with maximum conservation of the fibrous and ligamentous links.

Many operators have tried numerous techniques of ligamentoplasty, with variable results. It is also possible to rebuild the posterior band of the triangular ligament (Zancolli procedure).

The most difficult problems to solve are radius malunions with a DRUJ dislocation. According to the importance given to the limitation of the flexion–extension range or the blocking of supination, the choice can be either a corrective osteotomy of the radius malunion or a restoring operation of pronation–supination.

Generally, the corrective radius osteotomy is made to re-establish the normal orientation of the glena and the length of the radius, using a trapezoidal bony graft with a plate fixation. It is a difficult procedure because the exact calculation of the graft must take in account three factors—the length, the correction of the posterior tilt and the correction of the lateral shift. The fixation of these bony pieces with a plate is also difficult because of the lengthening; it is easier with an external fixator. It is also possible to make a cuneiform

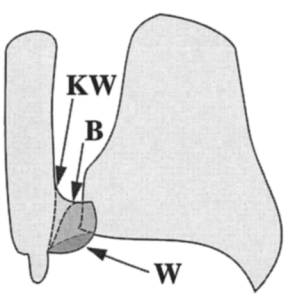


Figure 24. The shortening of the ulnar head. Three procedures may be used for caring for a painful impingement of the ulnar head: Two open procedures: Kirk Watson (KW) and Bowers (B), and one arthroscopic: Wafer procedure (W).

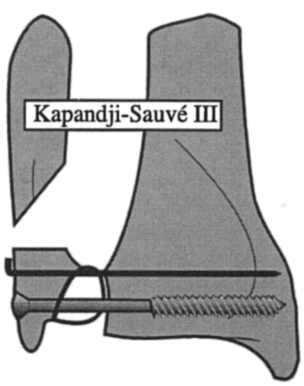


Figure 26. The Kapandji–Sauvé III procedure. After peeling the DRUJ cartilages without tilting the ulnar head, fixing it to the radius with a cancellous screw and a pin, then resecting obliquely the ulnar shaft to let a minimum gap of 5 mm in pronation position.

resection of the radius so as to reorient the glena, but this provides a shortening that must be compensated on the ulna, with a shortening, for example.

When the lack of supination is considered as the heaviest handicap, two procedures can be used: avulsion of the ulnar head (Moore–Darrach) or the Sauvé–Kapandji procedure (fusion of the DRUJ associated with a segmental resection of the lower ulnar shaft). This operation gives a rapid and excellent result in recovering supination, but, in its original version, it needed a complete tilt of the ulnar head so as to peel completely the cartilage surfaces of the DRUJ, and a resection 15 to 20 mm long of the lower ulnar shaft. It now is recognized that this type of procedure was a factor in ulnar stump instability. Consequently, a new procedure was described (Kapandji–Sauvé III; Fig. 26). The joint cartilages are rubbed out through a forced widening of the DRUJ and, after fixation of the ulnar head in the sigmoid notch in medium position with a cancellous bone screw (sometimes two screws or an added Kirschner wire to avoid a rotation of the head around the first screw), an oblique resection of the lower ulnar shaft is done, without destroying the connection of the ECU sheath with the ulna. The oblique part of this cut is directed distally and medially, and it is important to check that, in pronation, the edge of this cut does not make contact with the lower cut. The gap in this position must be 5-mm wide. With this procedure, the ulnar stump is much more stable. When there is an impingement of the ulnar head to the carpus, the procedure may include lifting the ulnar head before fixing it in the sigmoid notch. This procedure may also be combined with corrective osteotomy of a radius malunion.

Finally, in some cases, a DRUJ prosthesis (Fig. 27) would be a good solution[4] for difficult problems, but this type of material is still experimental.

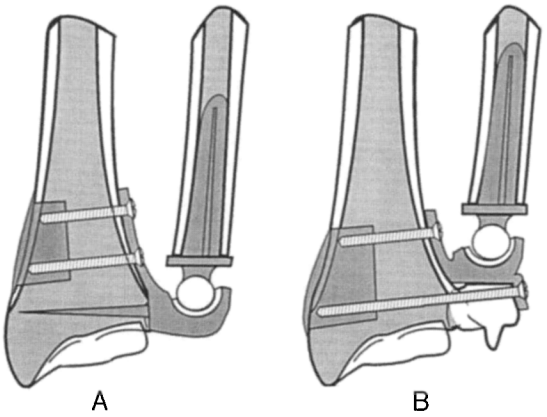


Figure 27. The DRUJ prosthesis. In an experimental process, a DRUJ prosthesis has been proposed with two types: without an ulnar head *(A)* after a Moore–Darrach procedure and with a remaining ulnar head *(B)* after a Kapandji–Sauvé procedure with an ulnar stump instability.

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